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DISCUSSION PAPER

A revised framework for digital transformation strategies: Contemporary insights and future research pathways

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Abstract

Digital transformation is strategically important for organizations. To accomplish successful digital transformation, organizations must develop appropriate digital transformation strategies that serve as a blueprint and set the overall direction of their digital transformation. The widely cited digital transformation framework by Matt et al. (2015) underscores the role of digital technologies in transforming value creation and organizational structures within the context of an organization's financial situation. Over the past decade, organizations have gained considerable experience in developing and executing digital transformation strategies. Recent technological developments, particularly in artificial intelligence, and the discourse around the role of identity in digital transformation, necessitate a reexamination of this framework. In the face of these developments, we analyze three successful digital transformation cases to put the framework to the test. As a result, we propose a revised framework for digital transformation strategies. In particular, we ground the model in the logic of digital innovations and suggest identity as a new central dimension. Based on the revised framework, we discuss several opportunities for future research. Our work is relevant to the discourse on digital transformation strategies as well as to corporate decision-makers who oversee or engage in digital transformation initiatives.

Keywords Digital transformation strategies · Digital transformation · Digital innovation · Strategy · Case study · Business model · Organizational identity · Artificial intelligence

JEL Classification M150

Introduction

Digital transformation, a fundamental change process that organizations undergo as they systematically employ digital technologies (Vial, 2019), remains a key priority for senior executives (Thomson Reuters, 2024) and is an important research topic in digital business (Hess et al., 2024). Organizations develop and execute digital transformation strategies (DTS) that serve as a blueprint for accomplishing significant changes through digital technologies (Matt et al., 2015). Research has suggested several frameworks for DTS that aim to summarize the most essential concepts and their interrelationships. These frameworks operate at various levels of abstraction. The DTS Framework by Matt et al. (2015) strikes a balance between generalizability and specificity and has received substantial attention from research. The framework suggests that DTS must consider the impact of the far-reaching use of digital technologies on changes in value creation and structural changes against the backdrop of available financial resources. Hess et al. (2016)

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further specified this framework to offer options for organizations to formulate DTS.

Over the past decade, there have been considerable developments in research and practice that should be considered for the development and execution of DTS. First, practical developments require revisiting and updating the digital transformation framework. From a technological perspective, artificial intelligence (AI) has emerged as a general-purpose technology that challenges companies' business. This raises the question of whether and how organizations adapt their DTS to AI. Second, the research discourse on digital transformation and its underlying strategies is vivid. One aspect that has recently received increased attention is the role of identity in digital transformation (Graf et al., 2023; Wessel et al., 2021). However, identity has not yet been considered within frameworks for DTS. Finally, a recent editorial called for a more holistic examination of digital transformation (Wessel et al., 2025). Taken together, these arguments call for a reexamination of the DTS Framework, leading to the following research question:

> What core components constitute a contemporary digital transformation strategy framework, and how are they interrelated?

This paper revisits important trends in the context of DTS and updates the DTS Framework by Matt et al. (2015). To this end, we conducted a multiple case study (Eisenhardt & Graebner, 2007) with Ringier, Otto, and Trumpf, all of whom are successful in their digital transformation and assign strategic priority to their digital resources. We interviewed senior executives of each organization and abductively (Dubois & Gadde, 2002; Timmermans & Tavory, 2012) analyzed the data against the original framework by Matt et al. (2015) as well as contemporary literature on digital transformation. Based on our analysis, we propose the DTS 2025 Framework. Compared to the original framework, the revised framework provides a more fine-grained differentiation of technologies and positions them as key drivers of digital transformation. It also newly integrates the dimension *role of identity* (Graf et al., 2023; Wessel et al., 2021) as an essential aspect of DTS. Reflecting on a more contemporary understanding of DTS, we further respecify several categories of the framework and provide several directions for future research.

Research background

Digital transformation strategies

The term digital transformation describes the use of different types of digital technologies to bring about substantial changes in the properties of an organization (Vial, 2019). Organizations employ novel digital technologies to create innovations that fuel digital transformation (Wiesböck & Hess, 2020). Specifically, innovative digital technologies provide a foundation for organizations to exploit innovative digital solutions and create innovative digital business concepts. Innovative digital solutions and business concepts must be tightly aligned to address different aspects of an organization's processes, products, or business models (Wiesböck & Hess, 2020).

Owing to the scope and far-reaching consequences of digital transformation (Vial, 2019), organizations develop DTS to prioritize, guide, and coordinate the many interrelated aspects of digital transformation (Hess et al., 2016; Matt et al., 2015). In contrast to related concepts such as IT strategy (Bharadwaj et al., 2013) and digital business strategy (Teubner, 2013), a DTS is a blueprint that organizations develop to steer activities related to digital transformation on a strategic level (Matt et al., 2015) and has been regarded as an important prerequisite for successful digital transformation (Gurbaxani & Dunkle, 2019).

Frameworks for DTS can be sorted according to their level of abstraction. First, the typologies that offer classifications of DTS have the highest level of abstraction. Second, there are frameworks that describe the general content of DTS, irrespective of the industry or an organization's market offerings. These contents include, for example, the organizational structures that organizations develop to support digital transformation and the financial frame in which they operate (Matt et al., 2015). Third, the research developed frameworks for specific industries, such as construction (Bhattacharya & Momaya, 2021). Table 1 provides an overview of the different frameworks used for DTS.

Table 1 Frameworks for digital transformation strategies

Level	Type	Example references
Level 1 – Highest level of abstraction	Typologies for digital transformation strategies	Fischer et al. (2020); Tekic and Koroteev (2019)
Level 2 – Medium level of abstraction	General frameworks for contents of digital transformation strategies	Hess et al. (2016); Matt et al. (2015)
Level 3 – Lowest level of abstraction	Industry-specific frameworks for contents of digital transformation strategies	Bhattacharya and Momaya (2021); Cipriano and Za (2024); Trenkle (2019)

Research on the typologies of DTS suggests that companies can adopt one of the four general types of DTS depending on their use of technology and how readily their business models can be digitally changed (Tekic & Koroteev, 2019). Thus, an organization can pursue a disruptive DTS if both technology use and business model readiness are high, and a technology-led strategy when technology use is high, but business model readiness is low. If technology use is low, an organization can either pursue a business-model-led DTS when business model readiness is high or be proud to be analog when business model readiness is low. While these types help organize DTS, they remain abstract and difficult for practitioners to operationalize. For example, the components that should be considered when developing such strategies remain unclear. Further, the strategy archetypes for digital transformation by Fischer et al. (2020) are based on the six core elements of business process management (BPM) and are at a similar level of abstraction. Additionally, they do not account for the fact that, in times of digital transformation, BPM follows a new logic (Baiyere et al., 2020).

Several frameworks address the *contents* of DTS. Many of these frameworks are tailored for specific industries. These frameworks are placed at the third level of abstraction. For example, there are frameworks for DTS of NGOs (Cipriano & Za, 2024), construction (Bhattacharya & Momaya, 2021), and small and medium-sized enterprises (Trenkle, 2019). Although these frameworks have the advantage of being tailored to one specific industry, they lack generalizability.

A DTS framework that strikes a balance between generalizability and specificity and is well received in research is the digital transformation framework proposed by Matt et al. (2015). This framework can be sorted into the second level of abstraction and is explained in detail below.

The 2015 Digital Transformation Strategy Framework

The digital transformation framework by Matt et al. (2015) breaks down the four dimensions of DTS: (1) financial aspects, (2) use of technologies, (3) changes in value creation, and (4) structural changes. Hess et al. (2016) further specified the framework to outline various categories along these dimensions as well as the associated options that companies can choose to formulate their DTS.

First, the framework emphasizes that digital transformation depends on financial aspects. This includes financial pressure and the financial resources available for digital transformation. Financial aspects can be both a push and roadblock for digital transformation. When a company is under severe financial pressure, it may lack opportunities to finance its transformation (Matt et al., 2015).

Second, digital transformation is influenced by the attitude and ambition of an organization towards new technologies (Matt et al., 2015). An organization needs to decide whether it wants to lead in terms of technology use or whether it wants to adopt established technologies and standards that already have proven business value for other organizations. While adopting technology early can lead to a first-mover advantage and force competitors to adopt self-developed standards, it requires technological expertise and is riskier. Likewise, adopting technologies late is less risky but might lead to missed opportunities and outperformance by competitors.

Third, technology use often leads to changes in value creation (Matt et al., 2015). Depending on the focus and extent of technology use, organizations may adapt their business processes (internal perspective) or existing products and services (external perspective). The external perspective captures the use of digital technologies to expand product and service portfolios. Moving away from the core business offers opportunities but is associated with higher risk and demands more nuanced competencies with regard to technology and products. This may also entail changes to the revenue model or even the business scope if new markets or customer segments are served.

Fourth, DTS deal with changes in the underlying organizational structure (Matt et al., 2015). This includes considerations regarding oversight and responsibility for digital transformation. For example, organizations may introduce the role of chief digital officer (CDO) to coordinate digital transformation or assign this responsibility to other personnel. Changes to organizational structure also include questions about how new competencies are acquired and whether new operations are integrated into existing structures or managed separately (Hess et al., 2016; Matt et al., 2015).

Two key observations necessitate a reexamination of the digital transformation framework. First, most research in the area of digital transformation focuses on a specific aspect of DTS without addressing the larger picture. For example, studies on CDOs (Singh et al., 2020; Tumbas et al., 2017) advance our understanding of this novel role but do not examine how the establishment of such a role might go hand in hand with changes in value creation, finances, and the use of new technologies. Because the different dimensions of DTS are interdependent, it is important to study them together.

Second, recent years have seen several developments in research and practice that are important for the formulation and execution of DTS. Since the introduction of the DTS Framework in 2015, new technologies have become increasingly popular. Most notably, AI has emerged as a general-purpose technology, and the extent to which the framework must be adapted to account for this is unclear. From an academic perspective, there is an ongoing discussion on the role of organizational identity in digital transformation (Graf et al., 2023; Wessel et al., 2021). It is

Table 2 Case companies

	Ringier	Otto	Trumpf
Industry	Media	Commerce	Engineering
Number of employees	7,000	6,000	17,000
Ownership	Family-owned	Family-owned	Family-owned
Headquarters	Zurich, Switzerland	Hamburg, Germany	Ditzingen, Germany

currently unclear whether digital transformation necessitates a change in organizational identity and how such a change might affect other dimensions of a DTS, such as value creation (Graf et al., 2023; Piccoli et al., 2024). Furthermore, while Wessel et al. (2021) highlight the relationship between organizational identity and value propositions, they do not consider other strategic aspects, such as an organization's financial situation (Matt et al., 2015).

The conceptualization of digital innovations (Wiesböck & Hess, 2020) offers a conceptual core that can be used to examine DTS. Against this background, we revisit the digital transformation framework by Matt et al. (2015), which we subsequently refer to as *DTS 2015*. Considering the developments in literature and three cases of successful digital transformation, we suggest a revised DTS framework and outline future research directions.

Research method

To examine contemporary DTS, we pursue a qualitative-abductive study design (Dubois & Gadde, 2002; Sætre and Van de Ven 2021) that is in line with prior studies on DTS (Hess et al., 2016). Specifically, we conducted multiple case studies (Eisenhardt & Graebner, 2007; Paré, 2004) of three organizations that have successfully undergone digital transformation. Case studies are ideal for studying such nascent phenomena (Eisenhardt, 1989; Yin, 2018), also because they allow us to dive deeply into individual cases to unravel their complexities in light of their idiosyncrasies. This section provides an overview of the cases and explains the data collection and analysis procedures.

Overview of cases

To select these cases, we employed purposive sampling based on three key criteria (Paré, 2004). (1) The cases should be successful instances of digital transformation because we wanted to learn how companies formulate successful DTS. (2) The cases should be family-owned businesses, as identity plays a large role in family-owned businesses and may be more easily influenced by owners (Wielsma & Brunninge, 2019). (3) The cases should be from different industries to understand how industry as a boundary condition influences the setup of DTS.

We selected the companies Ringier, Otto, and Trumpf, each of which is well-known in its respective industry. Table 2 presents an overview of the most important characteristics of the selected cases. All the cases are family-owned enterprises based in German-speaking areas (either Germany or Switzerland) that operate in many countries, if not globally. Trumpf is not only family-owned but is also currently family-run.

Ringier AG is a diversified media company headquartered in Zurich, Switzerland. Established in 1833, it evolved from a traditional printing business to a modern media and technology company with a broad portfolio of different media types and industries. Ringier is organized into three business areas: (1) media, (2) marketplaces, and (3) sports media. Over the last 15 years, Ringier has invested approximately two billion CHF in its digital transformation, primarily through mergers and acquisitions (M&A). Currently, approximately 83% of the earnings before interest, taxes, depreciation, and amortization (EBITDA) stem from digital activities. Ringier follows a decentralized management approach, granting the respective business areas entrepreneurial freedom to pursue their goals.

Otto GmbH & Co. KG, commonly known as Otto, is a major German e-commerce company and part of the Otto Group. Otto was founded in 1949 by Werner Otto in Hamburg, Germany. Initially, it was a mail-order company, starting with a hand-produced catalog featuring different styles of shoes, clothing, and household goods. Over time, it has evolved into one of the largest e-commerce companies in the world, operating in over 20 countries. To contrast it with the Otto Group, which includes various other businesses, we refer to Otto GmbH & Co. KG simply as Otto.[1]

The Trumpf Group is a German high-tech company known for its manufacturing solutions for machine tools, laser technology, and electronics. Founded in 1923 by Christian Trumpf as a mechanical workshop, Trumpf is still family-owned and family-run. The current CEO is Dr. Nicola Leibinger-Kammüller. Approximately 80% of Trumpf's customers are small family-owned businesses. To stay ahead of the market, Trumpf invests substantially in R&D. Over the last few years, R&D investments have accounted for

[1] https://www.otto.de/unternehmen/de/unternehmen/auf-einen-blick

approximately 9–10% of annual sales and are systematically higher than those of most competitors.

Data collection

We conducted in-depth interviews (Myers & Newman, 2007) with key stakeholders of the three organizations. Before conducting the interviews, we performed web research on each case organization to obtain preliminary and public information on their digital transformation. This included information from corporate websites, podcasts, white papers, and a TV documentary. Our case organizations consider their digital transformations successful, and their digital resources receive strategic priority (Piccoli et al., 2024). Our interview partners are regular speakers at industry events or universities who shared their learning experiences and reflections. For each case, we received an internal PowerPoint presentation summarizing the digital transformation. We collected all information pertaining to the different case organizations in the case database (Yin, 2018).

We further prepared the interviews by creating a semi-structured interview guideline (Myers & Newman, 2007). We used the questions devised by Hess et al. (2016) as a starting point and extended them in two ways. First, we incorporated additional questions regarding recent topics on DTS, such as identity (Graf et al., 2023; Wessel et al., 2021) and AI (Hanelt et al., 2021; Holmström, 2022). Second, to gain a longitudinal perspective (Langley, 1999) on how DTS unfold, we introduced new questions and modified existing ones. For example, we included a question outlining how the DTS has changed over the last 10 years. We also asked the interviewees to describe major roadblocks and how they were overcome, and explicitly asked for lessons learned. We provide the complete interview guideline in the appendix.

After collecting and analyzing publicly available information, we conducted three to four interviews per case. They include the CDO, CIOs, members of the management board, and a supervisory board member. Our thorough preparation and the presence of two or three members of the author team helped mitigate some of the challenges commonly associated with interviewing business elites (Ma et al., 2020; Solarino & Aguinis, 2021).

We conducted all interviews in German via Zoom and recorded them with the interviewees' consent. The interviews lasted between 45 and 76 min, with an average of 56:23 min. We verbally transcribed all interviews and added them to our case database (Yin, 2018). Table 3 summarizes the collected data.

Table 3 Data collection

	Ringier	Otto	Trumpf
Interviewees	• Head of Ringier Media • Head of Global Marketplaces • Chief Technology and Data Officer	• CIO • VP Strategy and Sustainability • VP Consumer Interactions • Head of Enterprise Architecture	• CDO • CTO • CIO • Member of the supervisory board
Access to internal documents on digital transformation strategies	Yes	Yes	Yes
Other documents	• Information from corporate website	• TV documentary • Podcasts • Information from corporate website	• Podcasts • White paper • Information from corporate website

Data analysis

Our data analysis followed an abductive logic (Dubois & Gadde, 2002; Timmermans & Tavory, 2012). That is, we coded all interviews against the body of knowledge on digital transformation (Vial, 2019) and corresponding DTS (Hess et al., 2016; Matt et al., 2015) as well as related concepts. In particular, we used the existing DTS 2015 Framework as a starting point but infused contemporary literature and concepts emerging from our data to extend the framework and introduce new concepts where appropriate.

All interviews were coded by one author using the Atlas.ti software. Throughout the entire project, we held regular meetings within the author team to discuss different project-related aspects, including data analysis and coding. This helped us corroborate the data analysis and foster the development of the revised framework. We coded the data along the three steps of open, axial, and selective coding.

First, we created open codes for all interview data. While we approached the data with the literature on DTS (Hess et al., 2016; Matt et al., 2015) in mind, we were sensitive to other concepts present in our data. This coding phase yielded 182 open codes. Table 4 provides examples of open codes and quotes from the interviews as illustrative evidence.

Second, during the axial coding phase, we coded around the categories that emerged as important given our research question. We found five essential categories: (1) use of digital technologies, (2) changes in structure, (3) changes in business model, (4) changes in identity, and (5) financial aspects. Four of the five categories are closely related to the original DTS 2015 Framework but have been adapted and extended to improve the comprehensiveness and specificity of the framework. Specifically, the dimension *use of technology* remained stable and did not require adjustments.

Table 4 Examples of open codes

Open code	Illustrative data
Learning from prior transformations	"But I am firmly convinced that the lessons we learnt from the synchro transformation have prepared us very well for this digital transformation. In other words, transformation as a competence within the company." (Member of the supervisory board, Trumpf)
Maker company	"Our digital value creation is relatively deep. In comparison to other German companies, we didn't do many outsourcing projects between 2000 and 2010 […]" (CIO, Otto)
Growth through M&A	"[…] little was developed organically, but we really chose the way that in general is a bit faster. When buying, we partially really had a very, very lucky hand, especially with regards to Jobcloud. Back then it was still a well-running startup, and today it is a huge company." (Head, Ringier Media, Ringier)

For the dimension *changes in structure*, we revised one category to better reflect the current discourse on capabilities for digital transformation (Baiyere et al., 2024; Piccoli & Pigni, 2013). We further realized that our case companies not only reported changes in their value creation but also in other aspects of their business models. To account for this, we introduced the dimension *changes in business model*, comprising not only how value is created but also how value is proposed to customers and how it is captured (Schüritz & Satzger, 2016). Next, because interviewees deemed identity an important aspect of their DTS, we included *changes in identity* as a new dimension and specified it based on the literature and our data. Finally, based on our interview data, we inductively derived a new category for the dimension *financial aspects* that addresses the financial planning horizon of an organization. Table 5 lists the axial codes and their subcategories.

Third, in the selective coding phase, we developed a revised DTS framework based on a combination of insights provided by established frameworks, the current literature on digital transformation, and underlying strategies, as well as insights generated from the analysis of our three cases.

Findings

In this section, we present the descriptive findings of our multiple case studies. We explain the DTS of our three cases in the following categories: (1) use of digital technologies, (2) changes in business models, (3) changes in structure, (4) changes in identity, and (5) financial aspects.

Ringier: From media to portfolio company

Ringier is a family-owned company, rooted in the media industry. Among other things, they publish the boulevard magazine "Blick." Ringier realized approximately 20 years ago that further growth opportunities with respect to their business model (media) and geography (Switzerland) were limited. Since then, the owner family has provided approximately 2 billion CHF to finance Ringier's digital transformation. Primarily through M&As, Ringier developed into a portfolio company with the three business areas media, marketplaces, and sports media serving markets in Europe and Africa. Ringier focuses on small markets, locally differentiating its products and services to attain a top market position and capture high margins.

Table 5 Examples of axial codes

Axial code	Category
Use of digital technologies	• Technological triggers for digital transformation • Strategic importance of IT • Ambition of technology use
Changes in business model	• Value proposition • Value creation • Value capturing
Changes in structure	• Formal responsibility for DTS • Development and acquisition of capabilities • Structural integration of new operations
Changes in identity	• External image • Self-image • Culture
Financial aspects	• Financial situation at the outset of digital transformation • Financing of digital transformation • Financial planning horizon

Use of digital technologies. Ringier's digital transformation was triggered by the rise of the Internet, which allowed it to offer online articles next to print issues. Because Ringier made substantial M&As and the business areas had a lot of entrepreneurial freedom, the management of technology in recent decades has been largely decentralized. Ringier is currently pushing towards more harmonization of technological and data infrastructures to leverage synergies and allow data sharing between different business areas.

Changes in business model. Although Ringier still publishes traditional print media, its interfaces with customers are becoming primarily digital. Online media has an ever-increasing share; the business areas of marketplaces and sports media employ entirely digital customer channels. Ringier uses personalization to tailor content according to customer preferences. Today, the largest share of revenue stems from the marketplace business unit. Revenues from media are likely to further decrease as individual media destinations become less important owing to AI agents that bundle media content. Marketplaces and sports media are growing faster than the media business. Among other things, this is because customers have very high interaction and attachment to affiliated betting websites on which sports media are published. The business area sports media not only provides sports media but also sells its content management system (CMS) to external customers, thereby serving as a technology provider.

Changes in structure. Owing to its portfolio structure, Ringier does not have a central CDO or strong CIO to oversee digital transformation. At the group level, a chief technology and data officer (CDTO) manages the technology platform shared across the organization. Owing to the ongoing harmonization of the technology and data infrastructure, the CDTO is gaining more authority.

Over the last 20 years, Ringier has substantially invested in M&A and has built strategic partnerships with companies that they consider friends and enemies ("frenemies") as well as with globally operating financial investors. Ringier followed a "buy and build" approach, which means they buy small companies to develop them into larger ones. As the market is consolidated, identifying attractive targets for making substantial profitable investments has become increasingly challenging. Consequently, Ringier is striving for more organic growth in combination with smaller "add-on acquisitions" to acquire specific know-how or technologies.

Changes in identity. Ringier moved further away from its original identity as a publisher and has, at least in part, become a technology provider. However, the publishing business remains an important part of the company that the owner family wants to sustain, as indicated by a school for journalism that Ringier operates.

With respect to culture, interviewees reported that Ringier has an open and innovative culture. Specifically referring to AI, it was noted that one needs to be careful about how to position its use, as a strong focus on headcount reduction can trigger employees' fear of AI, especially considering that Ringier has been cutting personnel costs in the recent past.

Financial aspects. Since the mid-1990s, the owner family has provided approximately 2 billion CHF for Ringier's digital transformation. The owner family has a long-term orientation that supports Ringier's successful development. The financial situation in the traditional media business has become structurally challenging; economically, business areas marketplaces and sports media are gaining importance.

Otto: Now a maker company

Otto's digital transformation occurred in two stages. First, Otto transitioned from a linear catalog commerce company to an online retailer. In 2008, Otto received 50% of its orders via catalog and 50% online. In 2015, this ratio increased to 90% online bookings.

However, Otto soon realized that this business model was only scalable to a limited extent and initiated the second stage of its transformation, in which Otto transitioned from an online retailer to a platform organization. As such, Otto sells not only its own products but also products from other companies and even competitors. In 2015, when the second stage was initiated, Otto offered 1 million products on its website. In 2021, 2000 partners had already joined the platform, which together generated approximately 1 billion euros in turnover. In contrast, the famous Otto catalog was discontinued in 2018, as it only contributed 0.3% of total sales at that point. Today, approximately 7000 partners operate on the Otto platform, which together generate approximately 3 billion euros in turnover.

Use of digital technologies. The Internet triggered the digital transformation of Otto into an e-commerce and then into a platform company. Our interviewees referred to Otto as a "maker" company that has deep digital value creation and develops most digital products itself. In contrast to many other companies, Otto did not conduct large outsourcing projects at the beginning of the 2000s. The platform was developed on a brownfield next to existing IT infrastructure. Otto works with agile methodologies and uses minimum viable products for software development that can be connected to the existing infrastructure in a modular fashion. Thus, new features can be easily tested, extended, or turned off, depending on their actual use by customers, as indicated by A/B tests.

Otto has established a portfolio process to oversee and align its software development activities. Overall, Otto's portfolio comprises 150 digital products developed and maintained by teams of 10 developers. Every quarter, business and IT executives discuss the complete digital portfolio and the work packages that should receive priority for the

next 3 months. Each project is managed by a triad of technical designers, business owners, and process owners, who steer the project from different perspectives. Otto operates two large software development hubs in Spain and India to enable 24/7 development. Otto invests approximately 10% of its IT budget in AI. There are already a variety of AI applications in different domains, such as analyzing product sustainability and creating product descriptions.

Changes in business model. With its move from a catalog commerce company to a platform provider, Otto's customer interface became completely digital. Otto generates revenue in three ways. First, Otto sells not only its own products but also competitors' products for commission. Second, in 2020, Otto acquired a banking license that allowed them to lend money not only to customers buying their own products but also to customers purchasing products from third-party providers. Third, Otto sells advertising spaces to customers to promote their products. Traditional merchants still account for approximately two-thirds of the revenue; however, other business areas are growing more quickly.

Changes in structure. Since 2015, Otto has had a strong CIO, also responsible for digital transformation. Along with the CEO, CFO, and the head of purchasing, this group forms the board of executive directors, which reports to the supervisory board. This is considered vital to Otto's transformation. Under the leadership of the CIO, the number of "technologists," comprising software developers, business analysts, data engineers, AI engineers, and similar job profiles, has doubled, constituting approximately one-third of the entire labor force at Otto. For large projects, Otto collaborates with Amazon, Microsoft, and other leading tech companies.

Otto has previously used innovation units to develop new capabilities. For example, to develop its banking capability, Otto created what they called a "capability organization," which was temporarily detached from daily business and later reintegrated into the organization. Temporal detachment makes it easier for involved employees to develop new expertise.

Changes in identity. For many customers, the printed "Otto Catalog" is the key symbol that represents Otto. Although the catalog is no longer printed, many people still associate Otto with it. This leads to a mismatch between Otto's identity as perceived by customers and employees, as Otto operates a digital platform business model.

All interviewees emphasized the importance of culture for achieving the digital transformation, and "organizational development" is a key strategy aspect, as everyone needs to pull along to achieve such a large-scale transformation. In this respect, it is difficult to change how an organization is steered. For example, for a long time, there have been KPIs in place to steer and incentivize employees to sell products. Changing the business model entails adapting these KPIs to ensure that each unit in the organization contributes to the same strategic goals. Furthermore, in the case of Otto, there have been cannibalization effects between third-party providers and Otto salespeople. This has led to frustration, and people had to be convinced that this change moves the entire organization forward.

Financial aspects. Otto is a privately owned company. The economic situation was stable, with constant growth at the outset of digital transformation. Because competitors such as Amazon grew faster, Otto lost market share. Thus, the key question for Otto was how to remain relevant and build a business model that allowed for long-term growth and economic success. Otto decided to transition from a traditional linear commerce model to a scalable digital platform business model.

Trumpf: Building digital services around high-tech machines

Trumpf initiated its digital transformation in 2017, triggered by an Industry 4.0 initiative by Fraunhofer and the upcoming migration of the enterprise resource planning (ERP) system to SAP S4/HANA. Their digital transformation can be divided into three stages. The first phase comprised the initiation of digital transformation. The second phase dealt with digital process excellence, and the third phase entailed the migration of its ERP system and accompanying business process management programs.

Use of digital technologies. From a technological perspective, Trumpf's most important move was the development of a reference architecture in 2018, which served as a digital platform. The platform was developed in collaboration with the consultancy Bain and a team of 20 employees from all the business units. It comprises several components that capture and integrate data from products, internal software applications, and customer-facing systems. It is connected to MyTrumpf, which is a central digital customer interface. The data collected allows Trumpf to improve its machine design and customer support. Despite the operational benefits, the development of the reference architecture was considered important for overcoming silos within Trumpf and establishing a standard for future software development projects.

Trumpf established a portfolio process for managing various software development projects. The portfolio is managed by the CTOs of the three business units and the CIO, who together form a governance board. In case of a disagreement, the CDO has the final authority. All decisions must be made according to the reference architecture, or the reference architecture must be adapted accordingly.

For Trumpf, software development is increasingly becoming a key capability. Over the past few years, their efficiency has increased by approximately 180%. This is possible

because of several strategic actions. First, most back-office IT was outsourced or near-shored to service centers to free capacity within Trumpf's corporate IT. Second, Trumpf globalized its IT by establishing three regional IT centers in Asia, America, and Europe. Third, they pursued a self-service approach, which allows business units to develop their own applications. Fourth, they improved requirements engineering and business case calculations to ensure that expected benefits can be realized. Fifth, Trumpf employed agile practices to accelerate the software development processes. Sixth, Trumpf strategically partnered with big IT players for foundational software development such as large language models and Industry 4.0 platforms.

Changes in business model. Trumpf primarily used digital technologies to change its value creation by improving business processes. In Trumpf's industry, there is little competition from digital business models. In addition, the monetization of digital solutions is difficult for Trumpf. Currently, digital solutions contribute approximately 160 million euros to the total turnover of 5 billion euros. However, these digital solutions may have additional value due to the lock-in effects on Trumpf's traditional products and services.

Trumpf has established a pay-per-part business model in which the machine is delivered to customers but operated autonomously by Trumpf. Customers can input the desired design and receive price quotations. If problems occur, there is a control center for remote support, or Trumpf sends physical operators to the customers. This business model has the advantage that customers no longer need to pay for the machine, and there is a price and quality promise for the envisioned parts. However, this business model has yet to diffuse widely.

Changes in structure. The CDO Mathias Kammüller oversees and is responsible for Trumpf's digital transformation. He is not only part of the executive board but is also married to the CEO Nicola Leibinger-Kammüller. He has been referred to as the "first change agent" at Trumpf and is a key figure for the digital transformation.

Trumpf pursues strategic partnerships with leading technology providers and consultancies to acquire competencies. They also collaborate with leading universities and research institutions to foster innovation. Additionally, Trumpf considers its close collaboration with customers a key asset for remaining innovative.

Changes in identity. Trumpf's identity has not changed significantly in recent years, as it is still primarily an engineering company. Digital solutions are considered add-ons, and customers do not seem to demand new business models, yet. Trumpf fosters an innovative culture to stay ahead of its competitors.

Financial aspects. Trumpf is a family-owned and family-run business. Although there are a few direct competitors, there is increasing pressure from cost-efficient Asian manufacturers. Trumpf must defend its price premium through higher machine quality and better service. In recent years, pressure from the owner family has increased, demanding a yearly growth rate of 10%.

A revised framework for digital transformation strategies

In this section, we suggest a revised framework for DTS that we refer to as *DTS 2025*. First, we explain the general dimensions of the framework before deriving the associated categories based on a comparative analysis of our three cases.

Dimensions of digital transformation strategies

The revised framework is illustrated in Fig. 1. It is based on the original DTS 2015 Framework. Based on the literature on digital transformation and DTS as well as our three empirical cases, we revised and extended the DTS 2015 Framework to provide an up-to-date overview of DTS for scholars and decision-makers. The DTS 2025 Framework encompasses five dimensions: (1) use of digital technologies, (2) changes in structure, (3) changes in business model, (4) changes in identity, and (5) financial aspects. Compared to the original DTS 2015 Framework, we changed the logic of the framework, incorporated new categories, and respecified existing categories to increase the specificity of the framework.

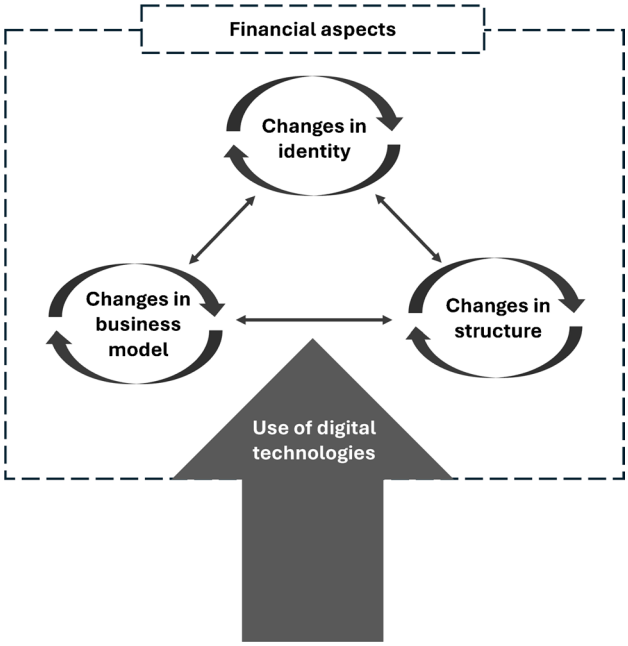

Fig. 1 The DTS 2025 Framework

The DTS 2025 Framework places the *use of digital technologies* as the driver of digital transformation. This dimension captures the technological perspective of DTS and includes technological triggers for digital transformation, the strategic importance of IT, and how ambitious the organization is with respect to the use of digital technologies. This positioning aligns with Wiesböck and Hess (2020), who argue that novel digital technologies offer opportunities for organizations to exploit and create digital innovations. This logic resembles the "technology-push," while organizations might also face "technology-pull" situations, where, for instance, changing business requirements lead them to use technology to fulfill these requirements. As digital innovations comprise product, process, and business model innovations, they are represented in the dimension *changes in business model* and are supported or constrained by *changes in organizational structure* and *changes in organizational identity*. Finally, *financial aspects* set the frame in which digital transformation and related activities take place.

Instead of changes in value creation, we now refer to the dimension *changes in business model* as it jointly comprises changes in value creation, value proposition, and value capture (Schüritz & Satzger, 2016). Value creation relates to *how* organizations perform their work. This perspective involves using different technologies to improve, change, and automate business processes (Badakhshan et al., 2022; Dumas et al., 2018). Further, the value proposition relates to *what* is offered to the customer in terms of products and services, and value capture addresses how a certain value proposition can be converted into *monetary remuneration*. What has previously been referred to as "future main business scope" in the DTS 2015 Framework is addressed in the value proposition, ensuring literature alignment and precision of the framework.

Digital transformation is often supported by *changes in the organizational structure*. This dimension encompasses the adaptation of existing roles and the creation of new ones to manage digital transformation and structural integration of new activities. We changed the former category of acquisition of competencies to *development and acquisition of capabilities*, as this labeling more accurately reflects the theoretical discourse on capabilities for digital transformation (Baiyere et al., 2024; Piccoli & Pigni, 2013) and more clearly emphasizes that these capabilities can be developed internally or acquired.

We further introduced *changes in identity* as a novel dimension in the framework that has been discussed in the literature (Graf et al., 2023; Wessel et al., 2021) and was prevalent in our data. Identity is a multifaceted concept. At a more granular level, the relevant aspects of digital transformation can be distinguished into external image, self-image, and organizational culture. First, "external image" refers to how an organization is perceived by its customers and competitors. For example, an organization's logo is a signal that allows customers to recognize the organization or brand. It should be noted that an organization can have multiple brands and that they might be perceived differently. Second, self-image captures how employees perceive their organization. This might be congruent with the external image; however, there can be substantial differences between the external and internal images. Third, organizational culture is the shared values of an organization that influence how its members act (Kummer & Schmiedel, 2016), and it has been highlighted as important for digital transformation (Hartl & Hess, 2017).

Organizational identity is not immutable but changes over time (Gioia et al., 2013). The use of digital technologies in the course of digital transformation can trigger changes in organizational identity (Graf et al., 2023), which are inherently tied to changes in the business model and vice versa. The more an organization departs from its original value proposition, the more the external image will change, as the value proposition is the key aspect of the business model with which customers are in touch. If the value proposition remains unchanged or relatively stable but value creation changes, the external image remains stable, whereas the internal image changes. The relationship between organizational identity and the business model is reciprocal; to fundamentally change their business model, organizations need to question and change their identity. Thus, we agree with Wessel et al. (2021) that digital transformation leads to a change in organizational identity; however, we argue that it can affect specific aspects of organizational identity in different ways. In the course of digital transformation, not all aspects of organizational identity need to change. For example, an organization's external image may be relatively stable, whereas its internal image may change substantially.

The DTS 2025 Framework includes *financial aspects* as its final dimension. Financial aspects frame DTS because they constrain or provide opportunities to invest in digital technologies and drive changes in other dimensions. Financial pressure increases the perceived urgency for digital transformation. However, once organizations struggle financially, they find it difficult to attract the financial means necessary to conduct digital transformation. Improvements in the financial situation may also be realized comparably slower than in other dimensions. While the organizational structure and business model can be changed over the short term, the financial situation is also dependent on the market in which the organization operates. Furthermore, a change in value capture can lead to changes in the revenue structure. By moving from a one-time purchase model to a subscription-based model, revenue is effectively shifted to the future. Decision makers must anticipate and reflect on this prior to changing the revenue model. This involves developing the capabilities necessary to operate the revenue

model (Gebauer et al., 2017). Furthermore, large investments accompanied by business model changes can lead to temporary financial setbacks. Thus, an organization needs to consider the temporal scale on which digital transformation needs to pay back (Tarakci et al., 2024).

Comparative analysis and derivation of categories

In our framework, the five aforementioned dimensions represent the highest level of abstraction, each capturing a fundamental area of decision-making relevant to firms. Within each dimension, we define a set of categories that delineate critical aspects requiring attention. To derive the categories, we conducted a comparative analysis in which we compared our three case companies along the dimensions of the revised digital transformation framework.

Use of digital technologies

For many media companies, including Ringier, the Internet was a trigger technology for digital transformation. As the quote below shows, the Internet also triggered Otto to pursue digital transformation. Thus, for both organizations, digital transformation began relatively early in the 1990s.

> "And the whole thing [Otto as a digital company] was created, because Michael Otto in 1995/96 came back from the United States and said: there is this Internet. […] In the late 2000 s, when the whole internet bubble stabilized, the share of orders that were placed only via the Internet and not via the catalog had increased dramatically." (CIO, Otto).

In comparison, Trumpf's digital transformation began much later, in 2017, when digital technologies became more sophisticated and were shown to be important for the manufacturing industry. Specifically, the digital transformation was triggered by the upcoming migration of their Enterprise Resource Planning (ERP) system to S/4 HANA, the emerging Industrial Internet of Things (IIOT), and the resulting need for data and software integration.

The strategic importance of IT for our case companies is high. They understood that digital technologies help them differentiate themselves from their competitors, and they make substantial investments in digital technologies that highlight their importance and enable their business models. For example, Otto operates an e-commerce platform that would not have been possible without digital technologies, and Trumpf uses digital technologies to complement machines and justify higher prices than its Asian competitors. The strategic importance of digital technologies is further highlighted by the fact that all case organizations have established development centers around the world to scale their software development capacities. The importance of business-led logic in the use of digital technologies is highlighted in the following quote:

> "[…] what do we actually need from a business perspective and how can technology help us? So technology is not an end in itself, […] but when in doubt, look at how we can best integrate our business into the technology […]. And then, in a very dovetailed way, with both perspectives, so to speak, then somehow look at it and think about how we can go into a future setup, so to speak, that perhaps, so to speak, in the conception, I say, then it's just about, when the target image is clear, then aligning the organization, and indeed the entire organization, precisely to this clear goal, so to speak, then in this second phase." (Vice President, Consumer Interactions, Otto).

The ambitions for technology use in our case companies vary. For example, all of our case companies explore how they can use AI, as they expect that it will have a fundamental impact on their business models. However, the ambition for use differs substantially. While all have developed use cases, Trumpf has formulated the most ambitious strategy for AI.

> "Digitalization and AI go hand in hand. As a leading provider of networked manufacturing, we at Trumpf also want to establish a market-leading position in the field of AI. For our customers, this means productivity and efficiency gains along the entire sheet metal process chain—regardless of the size of the company." (CEO, Machine Tools, Trumpf, as cited in Hönl (2024)).

Otto and Ringier can be characterized as early adopters, whereas Otto is slightly more innovative in the use of digital technologies. For example, they explore new technologies and conduct A/B tests to determine whether customers use new platform features. The use of technology at Ringier depends on the business area in question. In the business area media, the use of digital technologies is a defense against losses in market share. Their AI use cases mainly center on making journalistic work more efficient. By contrast, the business areas of sports media and marketplaces have state-of-the-art technical platforms that they sell to third parties.

Changes in business model

During their digital transformation, our case companies adapted their business models to varying extents. Trumpf streamlined its business processes (value creation) and complemented its machines with digital solutions but did not substantially change its value proposition. One reason

for this might be that their digital transformation started later than that of the other two case companies. Trumpf's CDO highlighted the role of digital solutions as a driver for machine sales but emphasized that there are no economically sustainable business model alternatives yet:

> "That was already the big question in 2017: will marketplaces, digital marketplaces, attack our business model? And that's not the case. So, there are a lot of them that have entered the market. None of them are successful, although they are well capitalized. But the system works quite well for normal parts, that you mill and spin and so forth. It's really difficult with sheet metal parts because you need a lot of customer interaction, especially with complicated parts, but with simple parts it works." (CDO, Trumpf).

Otto fundamentally changed its business model in two steps. The first shift, from a catalog commerce company to e-commerce, primarily changed value creation by changing the channel through which they reached customers. Its value propositions expanded through a broader assortment, whereas its value capture remained unchanged. The second shift, from e-commerce to a platform company, redefined value creation by adding banking and advertising capabilities. The availability of products further increased as they opened the platform for third-party sellers, thus impacting their value proposition. At the same time, value capture diversified, since they not only generated revenues from selling their own products but also received commissions from third-party sellers as well as from banking and advertising services. Consequently, Otto's traditional commerce business declined in importance, accounting for approximately 65% of the revenue, while the new revenue streams grew faster.

Also, Ringier's business model has changed considerably. With the rise of the Internet, they began offering articles online in addition to print issues, adding online advertising to their revenue streams. By acquiring different online marketplaces, they further extended their value proposition. In addition to advertisements, this line of business creates revenue through a subscription-based revenue model. Finally, sports media, while having the same revenue model as other media types, need to be updated more frequently, and thus, value creation is different.

Changes in organizational structure

With respect to organizational roles to support digital transformation, our case organizations chose different approaches. For Ringier, the heads of the three business units are responsible for managing the digital transformation within their business areas, following their entrepreneurial philosophy. They are supported by a group-wide CTDO, who has a supportive function of aligning different business units with respect to data and employed technologies. Otto has a strong CIO, who is also responsible for digital transformation and is a part of the management board. Finally, Trumpf has a very strong CDO who is not only part of the management board but also married to the CEO and, therefore, part of the owner family. In our interviews, the extraordinary role of Trumpf's CDO was highlighted. He took over the position of CDO, even though he had no track record with respect to the management of digital technologies.

> "I would say that Mr. Kammüller is the first change agent, […]so I think he has taken on the role of CDO from his family position. He has given himself the greatest point of instability in that he can present himself as a shareholder, namely, that you take over the act by not having a personal track record. But you have to be aware of the responsibility. I believe that all employees took this as a strong impulse and have continued to do so to this day. And I think he has constantly underpinned this with an excellent change, which ultimately manifests itself in the fact that you can call a member of the family by their first name. That's not the case with virtually all the other family members. And I think it was such a simple sign and perhaps quite normal for a modern company, but in the family context, it is for me. That what is visible to the outside world has to be dealt with, that different competence, different communication, that a different understanding of dialogue, that discourse at eye level is required." (CTO, Trumpf).

This diversity in terms of the roles in managing digital transformation shows that multiple configurations can contribute to successful digital transformations. However, all case organizations address digital transformation in their top management teams, giving this topic the highest importance and enabling its strategic management.

All case organizations emphasized that hiring qualified staff is becoming increasingly difficult, which limits their ability to develop competencies. Our case companies responded with individual solutions to this issue. Ringier follows a buy-and-build approach in which it buys smaller companies and develops them into larger ones. Otto and Trumpf scaled their software-development capacity by becoming more efficient. While Otto considers itself a "maker company," both Otto and Trumpf strategically collaborate with big players, such as Google and Microsoft, for large projects. The following quote by Trumpf's CTO illustrates this in the context of vision systems.

> "We developed vision systems by ourselves a decade ago, but we will not do this any further. So, in the future we will have partnerships, global partnerships

with companies, providing perfect vision systems, including specific vision AI sub models, for example. So, I think that the product structure of a machine tool builder will change more into an orchestrator of partners. And in the past, it was supplier relationships and integrated, and now it is developing in, I would say, a stringent management of systems engineering, including partners to provide solutions. That is the new normal in machine tools." (CTO, Trumpf).

As with the organizational roles in managing digital transformation and the development and acquisition of capabilities, the approach to the structural integration of new operations differs considerably across the three cases. Ringier does not integrate smaller companies but has previously created separate, autonomous business units. All investments in start-ups made by Trumpf are financial investments in independent companies that are not in direct competition with Trumpf products. Otto is the only one of our case companies that formed what they refer to as a "temporary organization" to develop competencies for banking. This unit was cut out and later reintegrated into the regular business. The head of Enterprise Architecture at Otto described this process as follows:

"[…] it was rather a temporary organization, I would say. So, there were ten or so business capability teams. For payment and marketplace and other such capabilities, which had the task of defining the new world for this hybrid platform model, so to speak, and then implementing the first steps. And after this went live with, let's say, this marketplace MVP, we then rolled this capability back or this organization back into the line. Or we set up a new line organization to some extent, because if I have a bank, I also need a line organization for the bank. We didn't have one like that, so you first had to free people up so that they could get out of the day-to-day business, enrich them a little with ideas from outside and then say, now you're also taking responsibility for the business behind it. That was roughly the procedure." (Head of Enterprise Architecture, Otto).

Changes in organizational identity

During digital transformation, an organization questions and adapts its identity. Our analysis highlights that not all aspects of organizational identity need to change. For example, Otto's external image has changed relatively little. This is because *what* Otto does (its value proposition) is, from the standpoint of its customers, relatively similar. However, *how* it conducts its business (i.e., value creation) has changed substantially. Thus, Otto's internal image has changed dramatically over the last 20 years. This discrepancy between the external and self-image is highlighted by a quote from the Otto CIO:

"[…] I'm always told that Mr. Müller-Wünsch from Otto is coming here with the catalog. These days, I usually leave the room and say that this has nothing to do with us anymore, because the catalog doesn't exist anymore since 2018 and we are now a digital company." (CIO, Otto).

Organizational identity, specifically the external image, is tied to customer expectations. For manufacturing organizations, high-quality standards for their products set expectations that products will be maintained in the future. In turn, these expectations make it more difficult for them to revise their respective identity. In other words, they are temporarily tied to their identity, which limits the space for possible changes to the business model. The following quote by Trumpf's CTO illustrates these long-term expectations.

"[…] we sign a contract with our customers, I would say, for decades. The machines that we deliver to our customers, they run for 15, 20, even sometimes 30 years, and they go into a very strong commitment with us. And we have to ensure that whatever we provide to the market can produce parts over decades." (CTO, Trumpf).

Another aspect of organizational identity is the culture that an organization exhibits and maintains. All case organizations emphasized the role of culture for digital transformation.

"What I think is most important is that we understand this as a change process, not just on the technological side, but also in terms of understanding people, so that we actually consciously take people with us. There are some fears. So, I'm seeing this here right now with AI and especially gen AI, where some people are saying, where is it going? Will it threaten my job or something? And it is therefore extremely important to not only have a good concept in terms of content, but also to get people excited and enthusiastic about it, to explain to people again and again what we are doing and why we are doing it. So you can't really underestimate the fact that you can only make the difference between success and failure in very, very large parts right at the corner." (Vice President Consumer Interactions, Otto).
"Digital transformation is first and foremost a culture topic." (Head of Global Marketplaces, Ringier).

Our analysis further highlights that all case organizations implemented agile approaches that are being infused into the organizational culture.

> "So, I believe you constantly do things that you then correct again. I think what was really important was that we started in the first place. And that we then constantly scrutinize and readjust almost everything we do. So that means we are constantly making corrections." (Head of Enterprise Architecture, Otto).

Financial aspects

With respect to financial aspects, all the case companies embarked on their digital transformation from a stable economic basis. At the same time, they anticipated that their financial situation would decline if they did not act. Owing to their good financial situation, our case companies were able to finance their digital transformation themselves and did not rely on external funding.

> "Of course, it's still a family business at the very end, and I'd say that's perhaps a little exaggerated, but quarterly results are important, but they can be overridden in terms of what the long-term goals are, and a family that has invested a good 2 billion in this first transformation. So, this willingness to further invest sustainably is really there. And yes, also the willingness to continuously pivot and adapt. I believe not every company has this kind of long-term [commitment]." (Head of Global Marketplaces, Ringier).

Long-term financial orientation and resilience to financial setbacks are likely to positively influence the success of digital transformation, as such companies are more reluctant to stop ongoing digital transformation if it does not lead to immediate financial results. Table 6 summarizes the DTS of our case companies. Figure 2 illustrates the DTS 2025 Framework, along with its associated dimensions and categories.

Table 6 Summary of digital transformation strategies of our cases

Dimension	Category	Ringier	Otto	Trumpf
Use of digital technologies	Technological triggers for digital transformation	Internet	Internet	Industrial Internet of Things; ERP migration
	Strategic importance of IT	High	High	High
	Ambition of technology use	Medium – High	High	High
Changes in business model	Value proposition	Media contents, information services	Products, platform for products, credits, and different services	Machines and complementary software
	Value creation	Content aggregation, fact checking	Development of platform to offer products	Improved and more efficient business processes
	Value capturing	Advertisement and subscriptions	Sales, commission, advertising, interests for consumer credits	One-time sales of machines and software
Changes in structure	Formal responsibility for DTS	CEOs of respective business areas and central CIO	CIO	CDO
	Structural integration of new operations	Integrated into daily operations	Creation of organization on time to develop banking capabilities	Portfolio-approach
	Development and acquisition of capabilities	Buy and build approach	Maker company	Strategic partnering
Changes in identity	External Image	Media company	Mail order company	Engineering company
	Self-image	Portfolio company	Modern platform business	Engineering company
	Culture	Innovative	Innovative	Innovative
Financial aspects	Financial situation at outset of digital transformation	Margin in media business declining	Good financial situation, but decline of market shares	Very good; pressure from low-cost manufacturers in Asia
	Financing of digital transformation	Self-financed	Self-financed	Self-financed
	Financial planning horizon	Rather long-term	Rather long-term	Long-term

The financial planning horizon of our case companies is relatively long. In this respect, our analysis suggests that family-owned businesses have an advantage, as they have a longer planning horizon and can handle temporary setbacks more easily than their stock-listed counterparts. One of the interviewees commented on this as follows:

Research outlook

Based on the revised framework, we suggest several directions for future research. We first discuss aspects concerning technology (Direction 1), followed by research directions regarding the business model (Directions 2a and 2b),

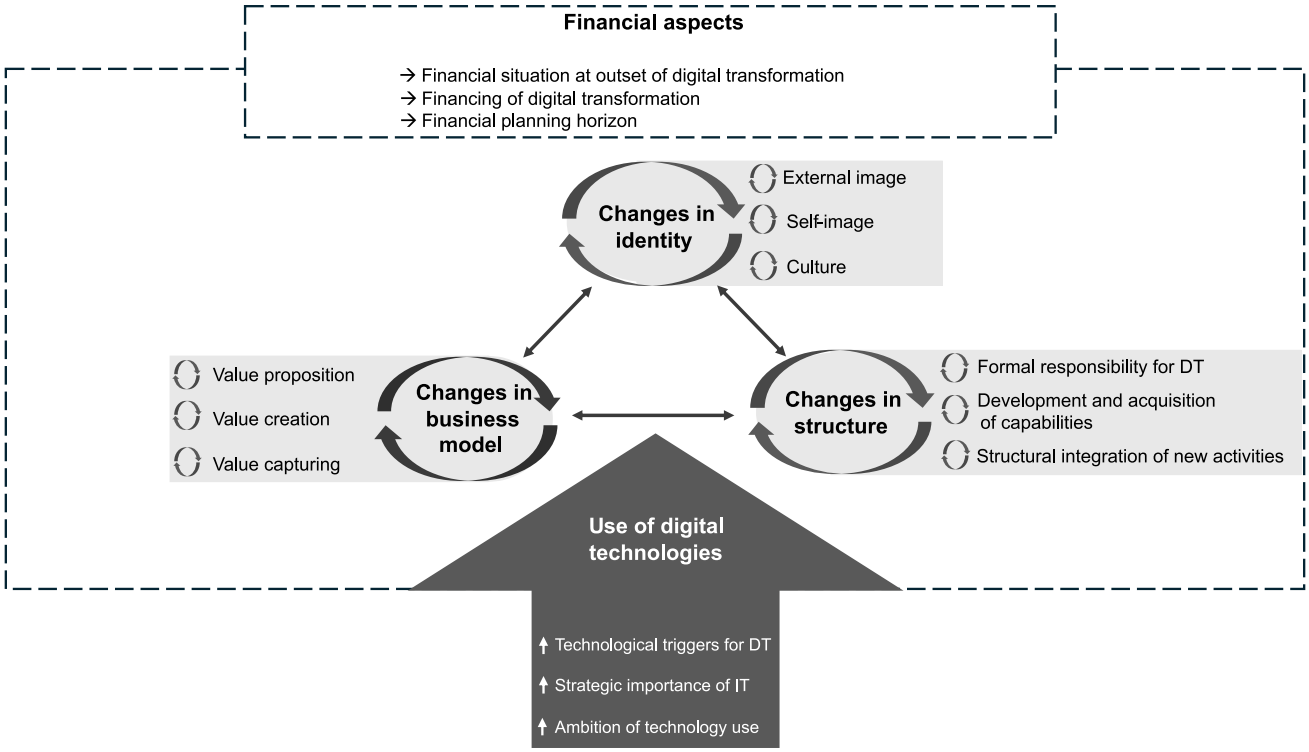

Fig. 2 The DTS 2025 Framework: Core dimensions and categories

organizational structures (Direction 3), organizational identity (Direction 4), and overarching observations (Direction 5).

Research direction 1: Towards a technology-agnostic framework

First, a crucial question is the extent to which the suggested framework is technology agnostic. One key motivation for our study was to understand whether novel technologies such as AI overthrow established frameworks for DTS. Based on our interviews, we found that this was not true. The dimensions of the framework appear to be sufficiently flexible for application across a wide range of digital technologies. However, our results indicate that AI creates a stronger urgency for digital transformation. All our cases investigate the extent to which AI can threaten their business model and how it can help them adjust their business model to stay relevant. For companies producing informational products, AI appears to be a greater threat. While manufacturing companies face more issues converting digital offerings into revenue, their core products remain unchallenged by AI.

However, two key questions remain unanswered. First, what parts of the framework are applicable to companies that have successfully undergone digital transformation and that are *digital* (Hovorka & Mueller, 2024)? Second, we need to test whether our suggested framework is applicable to entirely new classes of technologies and transformations. At this point, it seems plausible that quantum technologies drive the next wave of organizational transformations, leading to so-called "quantum transformations" (Piccoli et al., 2024). Future research should investigate the extent to which our framework is applicable to new classes of technology or whether entirely new approaches are necessary.

Research direction 2a: Pathways of expanding the core business

Second, when pursuing digital transformation, organizations need to decide whether and to what extent they want to expand their current core business. We suggest that organizations can employ at least four strategies to do so. First, organizations can offer the same or a very similar value proposition as in their analog business but via a digital medium. For example, in the first transformation phase, Otto developed from a print catalog retailer into an online retailer. Second, organizations can add services to existing products up and/or down their value chain. For example, Trumpf's core business is the development and production of machines. However, they added additional services that customers can use to design parts and improve machine maintenance. Third, organizations can augment their businesses. For example, Watkins Steel augmented its steel production business with services

for managing building sites (Chasin et al., 2022). This changed its position in the construction industry, as it not only provided building materials but also became information provider. Fourth, portfolio companies, such as Ringier, focus on the profitability of their business units. New business areas can be added in a modular fashion, and each unit operates with sufficient entrepreneurial freedom to pursue its goals. However, this comes at the expense of limited synergy among the different business units. Future research could further explore this preliminary differentiation to understand the circumstances under which each type is advantageous.

Research Direction 2b: product-type as a boundary condition for digital transformation strategies and their execution

We deliberately selected cases from diverse sectors to explore the role of industry context in digital transformation. Through our analysis, the relationship between analog and digital products emerged as a particularly important characteristic. The case of Trumpf, a leading global engineering company that produces machines and lasers, illustrates that physical products and resources can constrain digital transformation, particularly concerning the business model adopted by organizations. Services can be converted from analog to digital in a relatively straightforward manner. Therefore, media companies are generally considered forerunners of digital transformation and the general use of digital technology. Even then, organizations may struggle with the burden of technical debt (Rinta-Kahila et al., 2023). The modular design of digital technologies (Yoo et al., 2010) allows different software components to be combined through specified interfaces. In the case of physical products, this is more challenging because digital and physical resources need to be coupled, and physical resources do not have the same degree of modularity and connectivity as their digital counterparts. Furthermore, physical resources are not reprogrammable (Yoo et al., 2010) and thus cannot be modified as easily as digital resources. More research is needed to examine the properties of resources and how they affect the content and execution of DTS. To better understand this, we suggest further theorizing of physicality in digital innovations (Goebeler et al., 2024) as well as how this affects DTS. As with Research Direction 1, a qualitative-inductive study design seems to be the most appropriate for exploring these aspects in detail.

Research direction 3: New organizational structures to drive digital transformation

Third, an emerging body of work has begun to explore organizational structures that support digital transformation, such as digital innovation units (DIUs) (e.g., Barthel et al., 2020; Deist et al., 2023; Raabe et al. 2020; Schiffer et al., 2023). Among other things, the literature suggests typologies for DIUs (Barthel et al., 2020) and how companies can scale the digital innovations developed in DIUs (Schiffer et al., 2023). Interestingly, our case companies use different approaches to structurally integrate new activities. Trumpf does not operate separate innovation units; Ringier uses a portfolio approach that grants business units the freedom to pursue digital innovation. To develop entirely new capabilities, Otto had developed what they referred to as "temporary organization," with additional leeway for competency development, which was later reintegrated into their regular operations. Future research could further explore how companies can efficiently separate and reintegrate such DIUs. Additionally, research can explore how large-scale agile frameworks and New Forms of Organizing (NFOs) (Puranam et al., 2014) can help drive digital transformation. For example, research can investigate how organizations can employ Holacracy (Bernstein et al., 2016) and how it contributes to achieving digital transformation.

Research direction 4: The multi-faceted nature of organizational identity

Fourth, previous studies have highlighted the importance of organizational identity for digital transformation (Wessel et al., 2021). They found that digital transformation necessitates a change in organizational identity. We add to this line of work by observing that the malleability of organizational identity influences digital transformation to a considerable extent. It seems that the digital transformation of long-existing family-owned businesses is more difficult because there is a historic legacy that needs to be accounted for. The identity of the organization might be solidified, making it difficult for decision-makers to change or even question it. The larger and more diversified the organization, the easier it is to change its identity, as it has already changed in the past. Furthermore, our research suggests that, in addition to self-image and culture (Wessel et al., 2021), the external image (Dutton & Dukerich, 1991) should be considered during digital transformation. The external image and associated customer expectations are aspects of identity that remain underexplored. For example, Trumpf executives noted that their contracts with customers had been in place for decades. Thus, there is a strong commitment from both parties. These expectations temporarily limited the extent to which

Table 7 Future research directions

Research direction	Possible research questions
Direction 1: Towards a technology-agnostic framework	• How does the framework apply to novel classes of technologies? • Can the framework be applied to quantum transformations?
Direction 2a: Pathways of expanding the core business	• How do paths to extend the core business influence digital transformation? • When is one pathway preferable over another?
Direction 2b: Product-type as a boundary condition for digital transformation strategies and their execution	• How do physical properties influence modularity? • How can we theorize physicality in digital innovation and digital transformation?
Direction 3: New organizational structures to support digital transformation	• How do DIUs influence digital transformation? • How can DIUs effectively be carved out and reintegrated? • How can NFOs, such as Holacracy, be leveraged for digital transformation?
Direction 4: The multi-faceted nature of organizational identity	• How does organizational identity influence digital transformation? • How does organizational and IT legacy influence pathways for digital transformation? • How do customer expectations influence digital transformation?
Direction 5: Learning from other types of transformations	• How can insights from other types of transformations inform digital transformation? • How do strategies and principles from other types of transformations need to be adapted for digital transformation? • How is strategy-making for other types of transformations different from digital transformation?

Trumpf could change its business model. Future research should further examine the role of organizational identity in digital transformation. To do so, researchers should not only consider an organization's internal perspective but also external perspectives from customers or ecosystems (Ashrafi et al., 2024). A key question is, for example, how the historic legacy of an organization, considering different aspects of organizational identity, influences its ability to undergo digital transformation. To this end, longitudinal studies and process theorizing seem promising (Langley, 1999; Wiener et al., 2025).

Research direction 5: Learning from other types of transformations

Organizations can also capitalize on the experience from prior transformations for their digital transformation. For example, our interviewees at Trumpf reported that they profited tremendously from an earlier lean transformation led by the current CDO. Indeed, there is some overlap between the literature on lean and digital transformation. For example, Yadav et al. (2017) investigated the interrelationship between lean implementation and organizational transformation, and Al-Najem et al. (2012) examined the role of culture and leadership in lean transformation. Digital transformation research can be informed by research on other types of transformations, such as lean transformation. Thus, future research could explore how companies with prior experience in a specific type of transformation draw on this experience to guide their digital transformation. We should further explore what research has learned from other types of transformations and how this can be leveraged for digital transformation. In this respect, conceptual comparisons of different types of transformations and their impacts appear fruitful. Table 7 summarizes the outlined research directions and possible research questions.

Conclusion and limitations

The DTS Framework by Matt et al. (2015) has been highly cited and well acknowledged in the scientific community and strikes a balance between generalizability and specificity. Multiple developments in practice and academia over the last decade have mandated a revisit of the framework. Drawing on interviews with successful family-owned case companies, we developed the DTS 2025 Framework, which includes several adjustments and extensions of the original framework.

The DTS 2025 Framework suggests a change in the main driver of DTS. The revised model follows the logic of digital innovations in organizations (Wiesböck & Hess, 2020) and places digital technologies at the center. Financial aspects frame digital transformation, as they comprise the assets available to finance digital transformation, which is, at least in the long run, expected to contribute to the positive financial development of an organization. Second, the revised model includes both new and modified dimensions.

Notably, we introduce identity as a new dimension. We contribute to the literature on identity in the realm of digital transformation (Graf et al., 2023; Wessel et al., 2021) by conceptualizing identity in terms of external image, self-image, and culture. We further respecified the previous dimension *changes in value creation*, which we now refer to as *changes in business model* as a higher-level construct that also includes value proposition and value capture as categories. Introducing identity as a separate dimension adds to the comprehensiveness of the framework, while adjustments to other dimensions increase its specificity.

Our research has limitations stemming from the selection of the case companies. To propose a broadly applicable framework, we selected cases from different industries. However, all our cases have headquarters in German-speaking countries and are family-owned. Companies in other parts of the world may develop DTS with foci or priorities that are different from those of our case companies. For example, the recent AI Act passed by the legislative bodies of the European Union (European Commission 2024) is likely to affect the strategy of European companies regarding the use of AI. Second, family-owned businesses seem to have longer planning horizons than their stock-listed counterparts. This enables them to adopt a long-term perspective on digital transformation and invest where necessary. As a result of this long-term orientation, setbacks are less likely to lead to the abandonment of the DTS in these organizations. While the operationalization of DTS may differ, we contend that the dimensions of the DTS 2025 Framework are sufficiently general to ensure consistency across diverse geographic contexts and ownership structures.

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Declarations

Competing interest We have no competing interests to report.

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